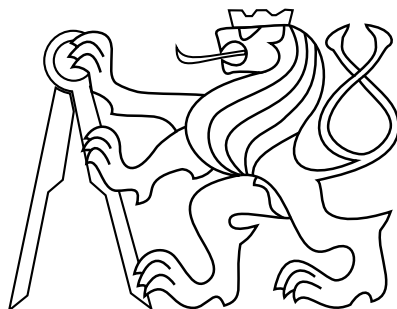


CZECH TECHNICAL UNIVERSITY IN PRAGUE

FACULTY OF ELECTRICAL ENGINEERING
DEPARTMENT OF CYBERNETICS



Classification of journalistic quality of articles

Bachelor's Thesis

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Study programme: Open Informatics
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Supervisor: Ing. Jan Šedivý, CSc.

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Swap this for the Thesis Assignment, when you have it from the department.

Declaration

I declare that this thesis was independently developed and that all sources of information used are cited in accordance with the university's ethical guidelines for thesis preparation. Additionally, I disclose that I utilized the AI tool ChatGPT to assist in translating certain sentences from my native language into English and to rephrase them in a style more appropriate for academic writing.

Prague, May 2025

Abstract

TODO:

Keywords Unmanned Aerial Vehicles, Automatic Control

Abstrakt

TODO:

Klíčová slova Bezpilotní Prostředky, Automatické Řízení

Abbreviations

TP True Positive

FP False Positive

FN False Negative

Class 0 Insufficient Article

Class 1 Short but Sufficient Article

Class 2 Standard Article

Class 3 High-Quality Article

GPS Global Positioning System

LiDAR Light Detection and Ranging

UAV Unmanned Aerial Vehicle

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1 Introduction

1.1 Motivation

The digital content revolution has transformed journalism into a high-velocity industry where platforms must simultaneously manage exponential content growth and maintain rigorous quality standards. Seznam.cz, serving over 5 million daily users, currently relies on human editors to manually assess only 20% of submitted articles - an unsustainable practice given the platform's 50% annual content growth rate. This manual process introduces significant challenges:

- **Scalability limitations:** With article submissions increasing by 50% annually, human editors cannot maintain pace without compromising review depth
- **Consistency issues:** Inter-editor agreement analysis reveals 20% variance in quality ratings for identical content
- **Quality degradation risks:** Unreviewed articles risk platform credibility through potential misinformation or poorly structured content

These limitations highlight the need for more efficient and consistent methods for evaluating journalistic quality. Automated quality classification addresses these challenges by providing:

- Real-time assessment for 100% of submissions
- Consistent application of editorial guidelines
- Significantly reduce the workload on human editors
- 24/7 operational capacity without human fatigue
- Data-driven insights for continuous quality improvement

Therefore, the primary motivation for this research is to develop an automated system capable of classifying the journalistic quality of articles, mimicking the complex decision-making process of experienced human editors.

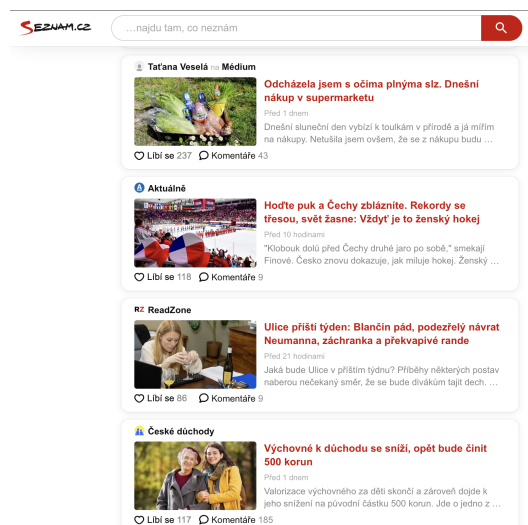


Figure 1.1: Main page Seznam.cz

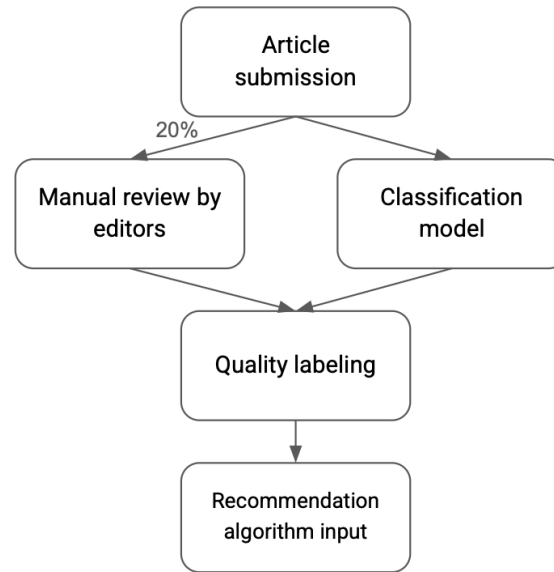


Figure 1.2: Current Editorial Workflow

■ 1.2 Problem Formulation

The core problem addressed in this thesis is the multi-class supervised classification of journalistic articles based on their quality. Our goal is to develop a computational model that automatically assigns an article to one of four predefined quality categories, reflecting the standards used by editors at Seznam.cz.

- Class 0 (Insufficient): Low-quality content that lacks structure and depth.
- Class 1 (Short but Sufficient): Lacks depth and analysis, often recycling known information with limited sources.
- Class 2 (Standart): Reliable news articles with at least five paragraphs and two references.
- Class 3 (High-Quality): Long, detailed, well-structured articles with visuals, subheadings, and references.

In the context of this thesis, **”journalistic quality”** is defined by a combination of factors that human editors consider. These include textual properties like readability, complexity, and style; structural attributes such as organization, paragraphing, and the use of headings; content characteristics like depth of analysis, factual accuracy, and source referencing; and metadata signals like the publishing channel. The aim is to create a model that learns to identify the patterns associated with these factors to predict the appropriate quality category. A significant challenge within this problem is the inherent imbalance in real-world data, where lower-quality categories are often much more frequent than the highest-quality ones.

■ 1.3 Structure of the Thesis

■ 1.4 Related Work

The automated classification of journalistic quality is a multidisciplinary challenge that integrates natural language processing (NLP), machine learning (ML), and media studies. This section critically reviews existing literature across these fields, highlighting significant contributions and identifying gaps that this thesis addresses, particularly in the context of the Czech language media environment and platforms like Seznam.cz.

■ Quality Assessment Frameworks in Journalism

Traditional journalistic quality assessment relies on human evaluation guided by professional standards. Lacy and Rosenstiel 2015 established eight core dimensions of journalistic quality, including accuracy, context, and sourcing diversity. While comprehensive, their framework was designed for manual evaluation, limiting its applicability to automated systems. Recent work by Calvo-Rubio and Rojas-Torrijos 2024 extends these principles to AI-assisted journalism, emphasizing transparency, accountability, and ethical considerations as critical criteria for maintaining quality in digital news production. Their study highlights that AI can enhance efficiency but risks reducing nuance and context, necessitating robust evaluation frameworks.

■ Computational Approaches to Text Quality Assessment

Automated text quality assessment provides foundational insights for journalistic applications. Pitler and Nenkova 2008 demonstrated correlations between syntactic features and human perceptions of writing quality, using metrics like sentence complexity and coherence. However, their approach was general and did not address the specific stylistic and ethical requirements of journalistic texts. More recent advancements, as noted in a 2024 article by Quantum 2024, leverage NLP techniques such as grammar analysis, sentiment analysis, and text cohesion metrics to evaluate content quality automatically. These methods offer promising tools for journalism but require adaptation to domain-specific standards.

■ Machine Learning for Classification of Media Content

Machine learning has significantly advanced media content classification. Potthast et al. 2018 developed methods to detect style-based bias in news articles, focusing on political orientation. Their work illustrates the potential of ML to identify subtle textual patterns but does not directly address overall quality. Similarly, González-Gallardo et al. 2021 applied BERT-based architectures to classify news articles by credibility, achieving accuracy rates above 85%. However, their binary classification approach (credible vs. non-credible) does not capture the nuanced spectrum of journalistic quality, which often requires multi-class models.

■ Multi-class Classification in Natural Language Processing

Multi-class classification in natural language processing (NLP) is a critical technique for categorizing text into multiple predefined classes, making it highly relevant for assessing the quality of journalistic content across various dimensions, such as accuracy, depth, and balance. A comprehensive review by Minaee et al. 2020 examines over 150 deep learning models applied to text classification tasks, including sentiment analysis, news categorization, and question answering. The authors highlight the effectiveness of transformer-based architectures, such

as BERT and its variants, which leverage large-scale pretraining and fine-tuning to achieve state-of-the-art performance in complex multi-class classification problems. These models are particularly promising for evaluating journalistic quality, as they can capture semantic nuances and contextual relationships within news articles. The review also summarizes over 40 popular datasets, some of which focus on news categorization, providing valuable benchmarks for developing quality assessment systems.

Similarly, Schmidt and Zangerle 2019 introduced a multi-class approach for classifying Wikipedia article quality using document embeddings and content features. Their method, which achieved high accuracy in distinguishing quality levels, demonstrates the potential of semantic representations for quality assessment, adaptable to journalistic contexts.

■ Linguistic Features for Quality Assessment

Identifying linguistic features that correlate with quality is a key research area. Schmidt and Zangerle 2019 utilized document embeddings to capture semantic content, combined with features like section length and readability scores, to assess article quality. In journalism, similar features—such as Flesch-Kincaid readability, lexical diversity, and syntactic complexity—can indicate quality but must be tailored to journalistic standards. The Czech language, with its rich morphology and inflectional system, presents unique challenges for NLP tools, as standard models designed for English may underperform.

■ Research Gaps

Despite progress, several gaps persist in the literature. First, most studies focus on English-language content, with limited exploration of languages like Czech, where morphological complexity affects NLP performance. Second, many approaches rely on binary classification, which oversimplifies the multi-dimensional nature of journalistic quality. Third, few studies address the specific needs of news aggregation platforms like Seznam.cz, where high content volume and diversity demand scalable, language-specific solutions. Finally, ethical concerns, such as bias in AI models and the loss of human judgment, remain underexplored in automated quality assessment.

This thesis aims to address these gaps by developing a multi-class classification model tailored for the Czech language media environment. By integrating textual analysis, metadata, and Czech-specific linguistic features, this work builds on existing research while tackling its limitations, particularly for news aggregation platforms.

■ 1.5 Contributions

This thesis makes several significant contributions to the field of automated journalistic quality assessment, particularly in the context of the Czech language and platforms like Seznam.cz. These contributions are outlined below:

■ Development of a Robust Classification System:

- Introduced a DeBERTaV3-based classification system for categorizing journalistic articles into four quality levels: Insufficient, Short but Sufficient, Standard, and High-Quality.
- Achieved a weighted F1 score of 0.77, closely approaching the human editor agreement upper bound of 80%, demonstrating the system's effectiveness in mimicking human editorial judgments.

■ **Innovative Handling of Class Imbalance and Overfitting:**

- Addressed the challenge of class imbalance (e.g., only 0.9% of articles are High-Quality) through sequential training, a hybrid loss function combining Cross-Entropy and Mean Squared Error, and class weighting ([1.0, 0.5, 1.0, 20.0]).
- These strategies effectively balanced model performance across all classes, particularly improving recall for the minority High-Quality class.

■ **Comprehensive Feature Engineering for Journalistic Quality:**

- Developed a feature set grounded in journalistic quality dimensions (Diversity, Transparency, Attractiveness, and Independence).
- Incorporated linguistic complexity features (e.g., syntactic complexity, lexical diversity), structural composition features (e.g., section organization, citation quality), readability metrics adapted for Czech, and metadata features (e.g., channel_id, publication timing).
- Utilized Latent Semantic Analysis (LSA) to reduce text embeddings to a single dimension, streamlining quality assessment.
- Introduced new features such as readability, manipulativity, and formality, addressing gaps in previous research.

■ **Comparative Analysis of Pre-trained Language Models:**

- Evaluated three pre-trained language models (BrowCzech, RetroMAE, DebertaV3 base and DebertaV3 large) for Czech journalistic content.
- Identified DebertaV3 large as the superior model due to its relative position embeddings and bilingual capabilities, providing insights for model selection in similar tasks, especially in low-resource languages.

■ **Advanced Model Testing and Superior Performance:**

- Tested multiple models, including a neural network with Bert-like models embeddings and a CatBoost classifier.
- Achieved the highest F1 score of 0.77 with the DebertaV3 large + CatBoost model.

■ **Innovations in Data Collection and Preprocessing:**

- Created refined datasets (e.g., editor-reviewed, unchecked) and cleaned text data by removing emojis and HTML tags.
- Highlighted the importance of article length as a quality indicator, with High-Quality articles averaging 1542.5 words compared to 486.5 for the majority class.

■ **Establishment of a Human Editor Benchmark:**

- Conducted a detailed editor agreement study, setting a realistic upper bound of 80% for automated systems.
- This benchmark provides a target for model performance and validates the complexity of journalistic quality assessment, ensuring that automated systems are evaluated against a feasible standard.

These contributions collectively advance the state-of-the-art in automated journalistic quality assessment, offering a scalable and effective solution for platforms like Seznam.cz while addressing technical challenges such as class imbalance, overfitting, and language-specific nuances. The thesis also lays a foundation for future research by documenting extensive experiments, baseline comparisons, and error analyses.

■ 2 Data Collection and Analysis

■ 2.1 Data Source

The data for this research originates from the internal systems of Seznam.cz, a major Czech online platform. The dataset comprises articles published on the platform, along with associated metadata and quality assessments provided by the editorial team. For the purpose of this thesis, we primarily utilized the following information for each article:

- **Title:** The headline of the article.
- **Content:** The full text body of the article.
- **Channel ID:** An identifier for the specific source or section within Seznam.cz where the article was published.
- **Entity type:** Type of entity (e.g., article, video, etc.)
- **Published date:** Date when the content was published
- **Content type:** Type of content (e.g., news, blog, etc.)
- **Annotated Quality Correction:** The quality label assigned by a human editor, serving as the ground truth for our classification task.

■ 2.2 Data Collection Timeline and Challenges

Data collection for this study began in July 2023 but faced significant challenges due to data transmission issues between departments. These issues affected the integrity of the dataset during the initial phases of collection. To illustrate the impact of these challenges, a plot (Fig. 2.1) of the number of labeled articles per day reveals inconsistencies in labeling frequency.

These problems were resolved after **August 30, 2024**, when accurate recording of confirmations began.

Additionally, discussions with editors revealed that the evaluation process evolved significantly over time. After **August 30, 2024**, more detailed training was introduced for editors, leading to cleaner and more consistent data. These improvements enhanced the quality of annotations, making the data collected after this date particularly robust.

As a result, the dataset used in this study benefits from a refined evaluation process and improved editor training, ensuring higher reliability and consistency in the annotations used for training and evaluating the classification model.

■ 2.3 Data Cleaning and Filtering

To prepare the dataset, the following steps were taken:

- (i) Articles with fewer than 50 words were excluded to ensure sufficient content for classification.
- (ii) Text was cleaned to remove emojis, HTML tags, and other non-text elements, focusing on title and main content.
- (iii) Only articles were included, excluding other entity types like videos or images.

These measures ensured a clean and relevant dataset for journalistic quality classification.

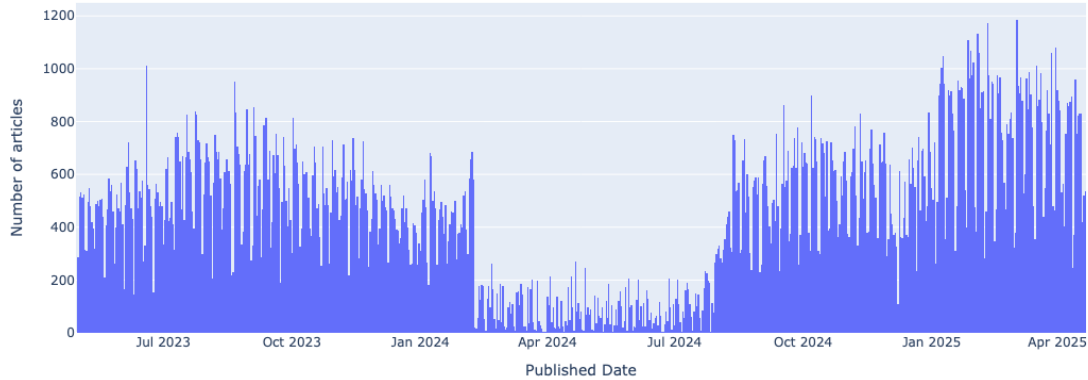


Figure 2.1: Number of labeled articles.

■ 2.4 Annotation Process

The quality labels used as the ground truth in this supervised learning task were assigned by experienced human editors at Seznam.cz. This annotation process is a routine part of content management on the platform, aimed at maintaining consistent editorial standards.

Editors evaluate articles based on a set of internal guidelines, but also rely significantly on their professional judgment and experience. They assess various aspects, including depth of content, writing quality, factual accuracy, structure, sourcing, and overall adherence to journalistic principles. Based on this assessment, each article is assigned one of the four quality labels defined in Section 1.2:

- Insufficient Article (Class 0)
- Short but Sufficient Article (Class 1)
- Standard Article (Class 2)
- High-Quality Article (Class 3)

■ 2.5 Dataset Statistics

■ Dataset Size and Splits

After cleaning and filtering, the data was split into three sets: Training Set, Validation Set and Test set. The distribution of articles across the four quality labels within each dataset split is presented in Tab. 2.1

Table 2.1: Class Distribution Across Dataset Splits

Set	Label 0	Label 1	Label 2	Label 3
Training	12,994 (22.9%)	32,086 (56.5%)	11,166 (19.7%)	542 (0.9%)
Validation	2,785	6,876	2,393	116
Test	2,795	6,863	2,404	118

As clearly visible from Tab. 2.1, the dataset exhibits significant class imbalance. The Class 1 constitutes the majority (over 56%) of the articles, while the Class 3 is extremely

rare, representing less than 1% of the data. This imbalance presents a major challenge for model training, as standard algorithms tend to become biased towards the majority classes. Addressing this imbalance effectively is a key focus of the methodology described in the next chapter.

■ 2.6 Data Analysis

■ Article length

Figure Fig. 2.2 shows the marginal distribution of article length in words for each class. The average article length is 545 words.

The distribution is strongly right-skewed: the shortest admissible texts contain 51 words, the median is 434 words and the longest article reaches almost 9000 words. This long tail already hints at a qualitative difference between routine news briefs and in-depth reporting.

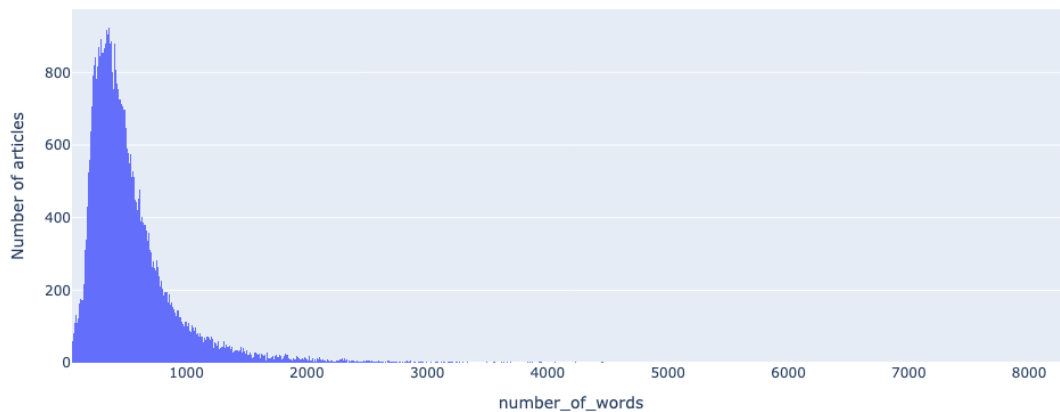


Figure 2.2: Distribution of article length in words.

■ Length by label

This Fig. 2.3 shows the distribution of article lengths for each quality label. The x-axis represents the number of words in an article, and the y-axis shows the probability density, so we can compare how article lengths differ across quality levels.

The results clearly show that better articles tend to be longer:

- Class 0 and Class 1 articles are the shortest. Most of them are under 600 words, with label 1 forming the sharpest peak around 300-400 words.
- Class 2 articles are longer, typically ranging from 400 to 1 200 words. They overlap with label 1 but clearly shift toward the right.
- Class 3 articles are the longest. They have a wide distribution, with many texts over 1 500 words and some even exceeding 7 000 words. The long tail in the purple distribution reflects very detailed, in-depth articles.

This confirms that article length is a strong signal for quality. Longer texts are more likely to include detailed explanations, sources, and structured content—all features editors associate with high-quality journalism.

However, since the distributions overlap, especially between classes 1 and 2, article length alone is not enough to classify quality.

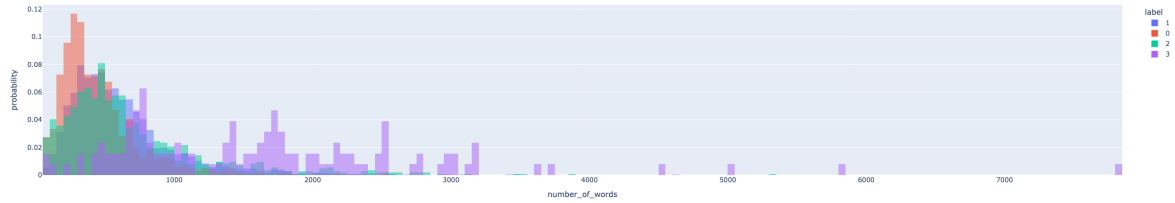


Figure 2.3: Distribution of article length in words by label

■ Unique-token probability density

The number of unique words in each article helps show how rich and varied the language is. Figure Fig. 2.4 shows that higher-quality articles usually have more unique words.

Class 0 often have between 280 and 380 unique words and rarely go above 800. Most articles in Class 1 also stay under 500 unique words. Class 2 articles use more words, and Class 3 articles often have over 1000 unique words. Some even go above 3000.

This means that better articles tend to use a wider vocabulary. However, there is still a lot of overlap between the groups, especially for shorter texts. So, the number of unique words alone cannot perfectly tell the article quality but is still a useful signal. It works best when combined with other features like article length and sentence structure.

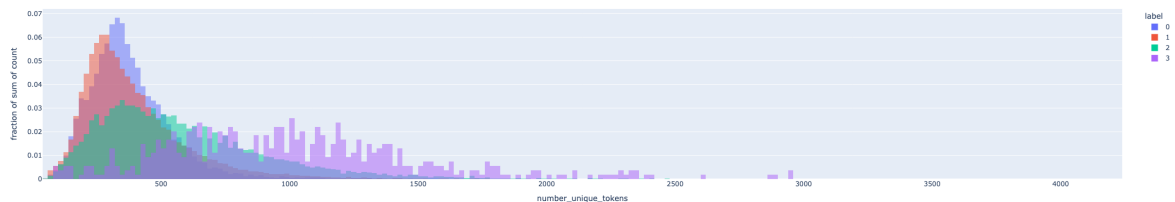


Figure 2.4: Distribution of unique tokens in articles.

■ 3 Evaluation Metrics

Evaluating the performance of a multi-class classification model for journalistic article quality requires careful selection of metrics, especially given the class imbalance in the dataset. As described in the Section 2.5, the four quality categories are not equally represented. This severe imbalance means that naive models could achieve seemingly high performance by favoring the majority class. Evaluation metrics must be chosen to reflect performance across all quality levels, rather than be dominated by the majority class.

■ 3.1 Limitations of Accuracy

Accuracy measures the proportion of correct predictions out of all predictions. While it is a simple and intuitive metric, accuracy is inadequate for imbalanced datasets. In our context of quality classification, accuracy treats all prediction errors uniformly and thus can be misleading. A model labeling every article as Class 1 would achieve 54% accuracy by virtue of Class 1's prevalence, potentially outperforming more nuanced models that try to distinguish all four classes but incur some mistakes on the majority class. This illustrates a key limitation: **accuracy can be artificially high even if the model entirely ignores minority classes.**

For a journalistic quality assessment system, such behavior is unacceptable – correctly identifying the minority classes (e.g. truly high-quality pieces) is crucial. Therefore, relying on accuracy alone would not highlight a model's failure to detect rare yet important cases. In light of this, we turn to alternative metrics that better capture performance across all classes, giving due weight to both common and uncommon article categories. These metrics focus on the balance between types of errors, rather than just the overall fraction correct.

■ 3.2 Precision

Precision is calculated as:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad (3.1)$$

where True Positive (TP) is the number of correctly predicted instances of a class, and False Positive (FP) is the number of incorrectly predicted instances where the model predicted the positive class but the actual label was negative.

Precision answers the question: “When the model predicts a certain class (especially a positive or high-quality outcome), how often is it correct?” In the context of our highest class (Class 3, high-quality articles), a high precision means that whenever the model flags an article as high-quality, it is usually correct. This is crucial for a platform like Seznam.cz because high precision in identifying top-quality content ensures that any article highlighted or promoted as “high-quality” truly meets the standards. In other words, high precision protects against false positives – it avoids promoting lower-quality articles as if they were premium content.

■ 3.3 Recall

Recall measures the fraction of actual positive instances correctly identified by the model, calculated as:

$$\text{Recall} = \frac{TP}{TP + FN}, \quad (3.2)$$

where False Negative (FN) is the number of actual positive instances that were incorrectly predicted as negative. It addresses “How many of the actual instances of a class did the model successfully find?” For Class 3, high recall means the model manages to identify most of the truly high-quality articles in the dataset. This is equally critical for our application: a high recall ensures that valuable content is not overlooked. In terms of business impact, high recall minimizes false negatives – it reduces the chance that an excellent article slips through the cracks and fails to be recognized or showcased as such. Missing a high-quality piece (a false negative) is a lost opportunity for user engagement and platform value, since exceptional content might not reach the audience it deserves.

■ 3.4 F1 Score

The balance between precision and recall is a classic trade-off: improving one can often degrade the other. In our case, both are important. To illustrate:

High Precision – Ensures that the articles labeled as high-quality are truly valuable, avoiding the risk of promoting subpar content to users.

High Recall – Ensures that most high-quality articles are detected and not ignored, maximizing the coverage of valuable content.

Depending on the application needs, one might prioritize precision over recall or vice-versa. For example, if the platform strictly wants to avoid ever featuring a low-quality article as top-tier content, it would emphasize precision (even if some good articles are missed). Conversely, if the goal is to surface as many great articles as possible, it would emphasize recall (even if a few lower-quality ones are misclassified among them). In this thesis, however, we aim for a balance, since both types of errors have costs (as discussed later).

To evaluate model performance holistically, we employ the **F1 score**, which is the harmonic mean of precision and recall. The F1 score provides a balanced measure that only yields a high value when both precision and recall are high, making it well-suited for our imbalanced multi-class problem. It is defined as:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3.3)$$

This formulation accounts for the trade-off between false positives (FP) and false negatives (FN): a model cannot maximize F1 by excelling in only one of precision or recall at the expense of the other.

Unlike accuracy—which, as noted, treats all errors the same and can be dominated by the majority class—the F1 score ensures that performance on minority classes is not masked. A high F1 indicates the model is catching most of the high-quality articles (high recall) and not mislabeling too many low-quality articles as high-quality (high precision). In practical terms, F1 was used as the primary metric during model development: for example, the training procedure monitored the F1 score on the validation set and employed early stopping when it ceased improving, ensuring the final model is the one that best balances precision and recall.

■ 3.5 Weighted F1 Score

When extending precision/recall/F1 to a multi-class setting, one must decide how to aggregate performance across classes. In this project we report the weighted F1 score as the primary evaluation metric, given the differing frequencies of each quality class. The weighted F1 is computed by calculating the F1 for each class separately, then taking a weighted average of these F1 scores using the class's support (proportion of samples) as weights. Formally:

$$\text{F1W} = \sum_{i=1}^n (w_i \times \text{F1}_i) \quad (3.4)$$

where $i \in \{0, 1, 2, 3\}$, $w_i = \frac{N_i}{N}$ is the fraction of samples belonging to class i in the dataset (i.e. its empirical probability) and F1_i is the F1 score for class i . This approach gives more influence to classes with more samples, while still incorporating performance on all classes.

■ Why use weighted F1?

In an imbalanced multi-class scenario like ours, weighted F1 offers several advantages over other averaging schemes (such as macro or micro averaging):

- **Balances Class Influence:** Macro-averaged F1 treats all classes equally, which can over-emphasize performance on very rare classes, while micro-averaging is dominated by the majority class (since it effectively counts every instance equally, making the aggregate metric similar to accuracy in a single-label setting). The weighted F1 is a compromise that considers each class's importance proportional to its frequency, preventing both the over-domination of majority classes and the over-sensitivity to tiny classes.
- **Comprehensive Single Metric:** The weighted F1 provides a single summary statistic that still reflects precision-recall balance across all categories. This is convenient for model selection and comparison. During development, we used the weighted F1 on the validation set to pick the best model and tune hyperparameters (see Methodology chapter on training procedure), since it ensured we were improving performance in an overall sense, not just for one favored class.
- **Reflects Real Data:** Weighted F1 aligns evaluation with the actual class distribution, ensuring that minority classes are considered without letting rare classes overly affect the result.

■ 3.6 Confusion Matrix

The confusion matrix is a powerful tool for visualizing the performance of a multi-class classification model. It provides a detailed breakdown of how many instances from each true class were predicted as each possible class. This allows us to see not only the overall accuracy but also where the model is making specific errors. Example of Confusion Matrix in Fig. 3.1.

In summary, the weighted F1 score was deemed the most appropriate metric for this study, and it is the primary metric reported for our results. It directly influenced model training as well: for instance, the final model was selected based on the highest validation weighted F1, and training was stopped early once this metric stopped improving to avoid overfitting. By using weighted F1 consistently, we maintain evaluation consistency between training and testing phases in the face of class imbalance.

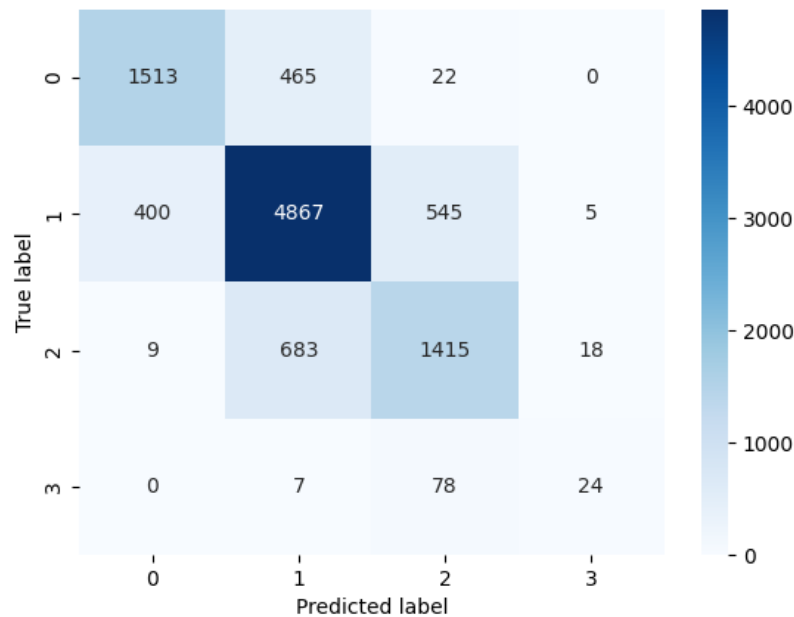


Figure 3.1: Example of Confusion Matrix. DebertaV3 Large.

■ 3.7 Business Relevance

The choice of evaluation metrics is not only a technical consideration but also a business decision in the context of a platform like Seznam.cz. The ultimate goal of automating journalistic quality assessment is to have the model’s predictions align with human editors’ judgments across all quality categories, from **Class 0!** (**Class 0!**) up to Class 3. As described in the Methodology chapter, human editors currently provide the quality labels, and their agreement rate sets an upper bound on model performance. The metrics we choose should thus reflect how well the model replicates those editorial decisions.

In this regard, the weighted F1 score aligns well with business objectives. It measures the model’s accuracy in mirroring human judgment while accounting for the prevalence of each quality category. In effect, a high weighted F1 indicates that the model is making the right distinctions between article qualities in roughly the same proportions as the real content mix on the platform. This is important for maintaining consistent quality standards: the platform seeks to reliably identify top-quality content and flag insufficient content, just as an editor would, but with far less manual effort. By using weighted F1 (which, as explained, ensures reasonable performance on each class), we directly support the goal that no class of article quality is neglected by the automated system. In business terms, this means the automation will uphold the overall editorial policy — it won’t simply label everything as “average” to get a good accuracy, but will correctly spot the outstanding pieces and the sub-par ones according to their true frequency. A metric aligned with these needs helps guide model development towards an outcome that is valuable for the platform’s content curation aims.

■ Cost of Classification Errors

In the context of content management, different types of classification errors carry distinct consequences, each affecting the platform in unique ways:

- **False Positives (FPs)** When lower-quality articles are incorrectly labeled as high-quality, they may be promoted as premium content. This can undermine user trust, as readers may feel misled by the platform's recommendations. Over time, such errors could harm the platform's reputation, making it less reliable in the eyes of its audience.
- **False Negatives (FNs)** Conversely, when truly high-quality articles are not recognized as such, the platform misses opportunities to showcase valuable material. This can reduce user engagement, as exceptional content fails to reach the audience it deserves, potentially limiting the platform's ability to retain and attract readers.

The F1 score addresses these challenges by striking a balance between precision (avoiding false positives) and recall (avoiding false negatives). This balance is crucial for managing the trade-off between promoting quality content and maintaining credibility. Additionally, the weighted variant of the F1 score ensures that less frequent categories—such as high-quality or insufficient articles—are not overlooked, despite their smaller presence in the dataset. By giving appropriate consideration to these minority classes, the metric supports a more comprehensive evaluation of the model's effectiveness, aligning with the platform's broader quality objectives.

■ 4 Methodology

- 4.1 Editor Analysis and Upper Bound
- 4.2 Baseline Model Performance
- 4.3 Performance of Pre-trained Language Models
- 4.4 Feature Engineering
- 4.5 CatBoost
- 4.6 Per-Class Performance Analysis
- 4.7 Error Analysis

■ 5 Experiments and Results

■ 5.1 Experimental Setup

Describe evaluation metrics Explain baseline models Detail hardware/software environment

■ 5.2 Model Comparison

Present results of different model configurations Compare performance across quality categories Analyze errors and confusions between classes

■ 5.3 Feature Importance Analysis

Identify most influential features Analyze how different feature groups contribute Discuss implications for journalistic quality assessment

■ 6 How to write thesis in LaTeX

■ 6.1 Versioning with git

Write the LaTeX in such a way that it could be versioned by git, which will help when collaborating with other people. This means writing **one sentence per line**. Even when you use third-party platforms, such as the OverLeaf, you can still share the repository through Git.

■ 6.2 Forming paragraphs

A paragraph is formed in LaTeX by an uninterrupted block of non-empty lines. It is recommended to keep a single sentence per line (helps with versioning using git). A new paragraph is started after an empty line.

This is a new paragraph. It is strongly recommended to **avoid** the use of the *newline* (`\\`) feature of LaTeX for forming paragraphs as it doesn't format the new paragraph properly (no space at beginning of the new paragraph).

■ 6.3 Linguistic anti patterns

■ Narrative

We recommend to write your thesis in plural form of the first-person narrative in combination with passive tense, e.g.:

- We discourage the use of any other form, and/or
- any other form is discouraged, but **not**
- I discourage you from using the first-person narrative.

Moreover, avoid “instructional” or “teacher”-like style of writing, such as “**Now, we multiply the matrix $\mathbf{A}$ by the scalar c to get the scaled matrix $\mathbf{B}$.**” A better way of writing the same information would be e.g. “Now, the scaled matrix $\mathbf{B}$ is obtained by multiplying the matrix $\mathbf{A}$ by the scalar c .”

■ Pronouns

The use of pronouns (it, this, they) is strongly **discouraged**. Although, pronouns make it easier for you as a writer to form the flow of the text, pronouns also make it much more difficult for the reader to follow the text. The reader is forced to retain more of the context to substitute and understand what the author meant. Moreover, pronouns can easily become vague (there is more than one way how to interpret them) and can become invalid while making editorial changes to the text, i.e., when moving sentences around. A technical text should be written in a way that makes it as easy to read and comprehend as possible and as hard to misunderstand or misinterpret as possible at the same time.

■ 6.4 Mathematical notation with LaTeX

Take care to use the correct mathematical symbols and common ways of denoting mathematical concepts. Use bold fonts to visually distinguish vectors and matrices ($\mathbf{x}$, $\mathbf{A}$) and

scalars (k , N).

■ Common errors

A frequent error, carried over from programming languages, is using the asterisk symbol ($*$) to denote multiplication. The asterisk correctly denotes convolution. Similarly, the cross sign ($\times$) typically denotes the cross product (it can also be used for stating dimensions, such as $10\text{ m} \times 10\text{ m}$) and thus should not be used for scalar multiplication. In English mathematical notation, **scalar multiplication is typically not denoted at all**.

This custom may sometimes make it unclear whether a sequence of letters denotes multiplication of several scalars or a multi-letter variable, such as

$$T = T0 + coef f meas, \quad (6.1)$$

where the variables in this hypothetical equation are T , $T0$, $coef$ and $meas$. For this reason, **avoid using multi-letter variable naming** and strive to denote mathematical variables with single letters optionally with a lower or upper index, or other modifiers ($\hat{}$, $\bar{}$, etc.). The equation above could be modified to be

$$T = T_0 + cT_{\text{meas}}. \quad (6.2)$$

If the multiplication is still unclear (e.g. when multiplying many single-letter scalars), the $\cdot$ symbol may be used such as

$$P \cdot V = n \cdot R \cdot T. \quad (6.3)$$

■ Equations

Mathematical equations should be numbered and should be a part of a sentence. For example, a discrete LTI system update is described as

$$\mathbf{x}_{[k+1]} = \mathbf{A}\mathbf{x}_{[k]} + \mathbf{B}\mathbf{u}_{[k]}, \quad (6.4)$$

where $\mathbf{x}_{[k]} \in \mathbb{R}^m$ is the state vector at the sample k , $\mathbf{u}_{[k]} \in \mathbb{R}^n$ is the input vector, $\mathbf{A} \in \mathbb{R}^{m \times m}$ is the main system matrix, and $\mathbf{B} \in \mathbb{R}^{m \times n}$ is the system input matrix. Proper punctuation should be used after the equation, as if it were an ordinary object in the sentence.

Do not put any empty lines before the equation. If the sentence that the equation is a part of continues after the equation (as is the case here), do not put empty lines after the equation either. That would create a new paragraph mid-sentence. **For an example of how not to do it, the equation**

$$\sigma(x) = \frac{1}{1 + e^{-x}} \quad (6.5)$$

describes the logistic function often used in machine learning. Observe how a new paragraph is created for the equation and then for this block of text (compare with the proper typesetting above). Not only does this not look correct, it may also cause incorrect page breaking.

■ 6.5 Using footnotes

Do not be afraid to use footnotes for additional information, such as http links¹. We use footnote links whenever we want to *point* to a website, rather than to cite it as a source. Like with everything, do not overdo it.

■ 6.6 Referencing document elements

LaTeX allows you to dynamically reference to parts of the documents, such as

- figures: Fig. 6.4, Figure 6.4,
- equations: eq. (6.4), (6.4),
- code: Lst. 6.1,
- and any other object that can contain a `\label`.

Check the section in the `document_setup.tex` that contains useful macros for unifying the references:

```
\newcommand{\reffig}[1]{Fig.~\ref{#1}}
\newcommand{\reflst}[1]{Lst.~\ref{#1}}
\newcommand{\refalg}[1]{Alg.~\ref{#1}}
\newcommand{\refsec}[1]{Sec.~\ref{#1}}
\newcommand{\reftab}[1]{Table~\ref{#1}}
\newcommand{\refeq}[1]{\eqref{#1}}
```

Listing 6.1: LaTeX macros for referencing to document elements.

■ 6.7 Abbreviations with Acronym

Abbreviations are handled by the *acronym* package. Example sentence with abbreviations: “Unmanned Aerial Vehicle (UAV) is a flying vehicle that commonly uses Light Detection and Ranging (LiDAR) and Global Positioning System (GPS) receiver”. Note that the acronyms are only explained once in the document by default. It is good practice to re-explain acronyms used both in the abstract and the rest of the document as the abstract is often presented separately. This can be achieved by resetting the internal status of the acronyms (“forgetting” that they were explained) using the `\acresetall` command after the abstract. Please, read the documentation².

■ 6.8 Units of measurements with Siunitx

Typesetting of units has never been more accessible with the Siunitx package. Acceleration is measured in ms^{-2} . Gravity accelerates objects at a rate $\approx 9.81 \text{ ms}^{-2}$ near the sea level. You can define your units if you want.

■ 6.9 Hyphens and dashes

Hyphens and dashes are the various form of the symbol “-” used in many situations. There are also various ways how to typeset the symbol in LaTeX.

- The *hyphen* is used to compound words, e.g., “the eye-opener”. The hyphen is typeset as a single *minus/hyphen* character: -.

¹This repository: <https://github.com/ctu-mrs/thesis-template>.

²Acronym package: <http://mirrors.ctan.org/macros/latex/contrib/acronym/acronym.pdf>

- The *en-dash* is used to specify ranges of values, e.g., “between 2–10”. The en-dash is typeset as two consecutive hyphens characters: --.
- The *em-dash* is used to separate complex sentences in place of commas, parenthesis and colons — each with its particular rules. The em-dash is typeset as three consecutive hyphens characters: ---.

Check the <https://www.thepunctuationguide.com/> for all the details.

■ 6.10 Double quotation marks

“Double quotes” in English are composed of a pair of opening (“) and closing (”) symbols. The opening symbol is typeset as two backtick characters: ‘‘ (typically below the Esc key on the English keyboard), and the closing quotes as two apostrophes: ’’. The LaTeX engine will convert them automatically to the opening and closing symbols. A more robust solution is to use the `csquotes` package and the `\enquote` command which also takes care of nested quoting and other peculiarities.

■ 6.11 2D Diagrams with Tikz

Tikz is a powerful tool for drawing 2D (and 3D) shapes and diagrams. Check the documentation and examples: https://www.overleaf.com/learn/latex/TikZ_package. The benefit of using *Tikz*, instead of some other third-party drawing program, are:

- fonts are the same as in LaTeX,
- you can typeset math in LaTeX,
- you can use references to other parts of your document,
- you can version the image in git,
- the images are easily adjustable while editing your document.

Check Fig. 6.1 for example.

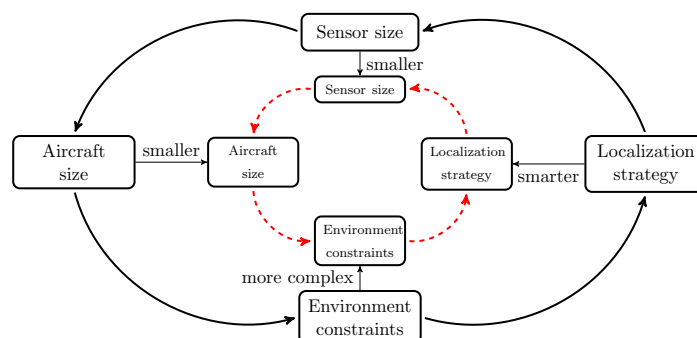


Figure 6.1: Example of a 2D diagram using tikz *PGFPlots*.

■ 6.12 Data plots with PGFPlots

PGFPlots produces nice 2D and 3D data plots from data stored in CSV. The plot parameters can be versioned and easily adjusted by editing the plot definition file.

- Documentation and manual: <https://ctan.org/pkg/pgfplots>
- Compile the plots individually and then include the pdfs because it can take longer.
- Example located in `fig/plots/example_plot`, see Fig. 6.2.

- You could include the latex file directly. However, it will take longer to compile, and platforms such as Overleaf can have a problem with that.

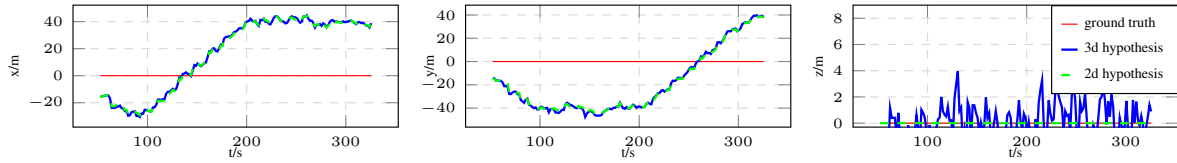


Figure 6.2: Example of a 2D plot using *PGFPlots*.

■ 6.13 3D Plots with Sketch

Sketch is a tool for defining a 3D scene using simple descriptive language. The 3D scene is then converted to *Tikz*, which is later compiled to pdf. The benefits of using *Sketch* are similar to using *Tikz*: LaTeX fonts, versioning using git, and cleanness of the result. See the example image in Fig. 6.3.

- Documentation and manual: <http://www.frontiernet.net/~eugene.reessler/>
- Cross-compilation from *Sketch* to *pdf* using the `fig/sketch/compile.sketch.sh` script.

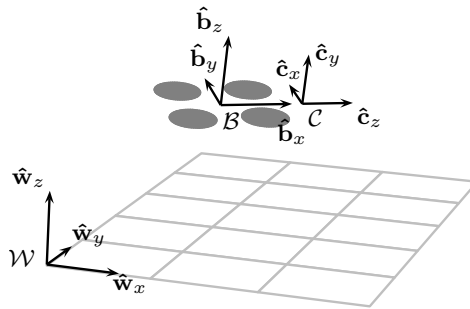


Figure 6.3: Depiction of the used coordinate systems. The image was drawn using *Sketch*.

■ 6.14 Image collages with Subfig

We recommend using the `subfig` package, which provides the `\subfloat` command. It is more versatile than the simpler `subcaption` package. Check Fig. 6.4 for an example.

■ 6.15 Citations with Biblatex

Biblatex is probably the most powerful citation package for LaTeX. It consumes the standard `.bib` file. However, it can sort and filter the citations using the `keywords` tag. Citing references is done using the `cite` command, e.g., Baca 2021. You can also define some nice citation boxes, such as this one:

Baca Baca, John (2021). “Some Title”. In: *Some Journal* 2021



(a) A UAV, the T650 model.



(b) Another UAV, again, the T650 model.

Figure 6.4: The caption should mention both subfigures, the Fig. 6.4a and the Fig. 6.4b. You can just refer to them as (a) and (b) in the main Figure’s caption, but beware, you need to keep it correct as you edit.

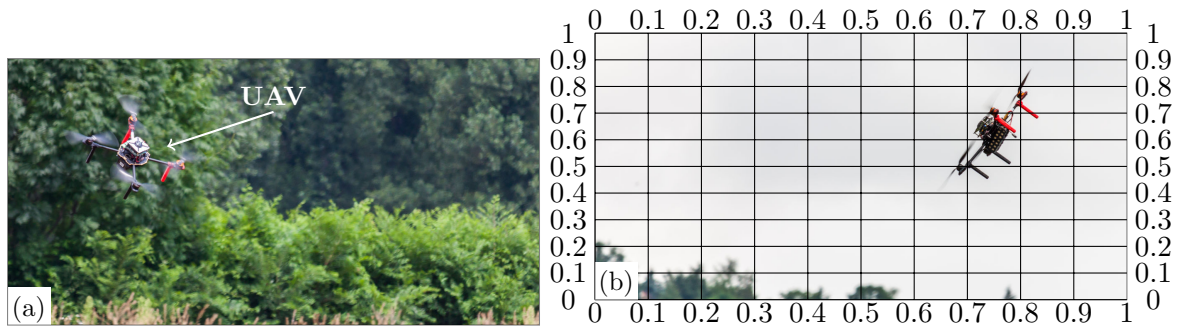


Figure 6.5: Example of using Tikz for image overlays. (a) shows a final product, (b) shows a grid useful for nailing down the coordinates.

6.16 Image overlays with Tikz

Tikz is very useful to create custom image overlays. The overlay can be set such that the image is spanned by Cartesian coordinates $(x, y) \in [0, 1]^2$. Example can be seen in Fig. 6.5.

6.17 General tips

In general, strive to make the paper easy to read and understand, and hard to misunderstand or misinterpret. Here are some more specific tips on how to achieve that (and other general suggestions).

- **Be consistent.** This applies in all contexts. For example, if you decide to use the name “LiDAR”, do not mix it with “LIDAR” or “Lidar”, do not mix different mathematical notations, ensure your Figures have the same style and use the same graphics for the same concepts, etc.
- After you finish writing or modifying any of:
 - a sentence,
 - a paragraph,
 - a section/chapter,
 - the whole paper/thesis,

re-read it to make sure that it makes sense, it is coherent and correct, and doesn’t

contain typos.

- If you're using a LLM-based tool (ChatGPT etc.) for grammar-proofing or even formulation of sentences, **do not just copy-paste its response** to your query. The previous rule applies doubly here. LLMs tend to often produce confident-sounding nonsense, sentences with reformulated duplicated content, or with a slightly changed meaning. They are a good tool to get inspiration to start writing about a subject, for grammar-checking, or for finding alternative, nice-sounding formulations, but they can lie or warp facts — take care when using them!

■ 7 Discussion, Conclusion and Future Work

■ 7.1 Discussion

Interpret the results in context of journalistic quality Compare your approach with existing methods Discuss practical implications for Seznam.cz

■ 7.2 Conclusion

Summarize key findings Revisit research questions and contributions Assess overall success of the approach

■ 7.3 Future Work

Suggest improvements to the model Propose additional features or approaches Discuss potential applications beyond the current scope

■ 8 Conclusion

Summarize the achieved results. Can be similar as an abstract or an introduction, however, it should be written in past tense.

9 References

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■ A Appendix A